INFINITELY MANY POINTS SPECIALIZE TO A NON-CLOSED POINT OVER AN INFINITE FIELD

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Decency and closed points. lemma-decent-space-closed-point (carded L26732). Closed points of the special fibre are dense.
- Jacobian criterion over infinite fields. Algebra, lemma infinite-field-polynomial (cross-chapter, named). Polynomials over infinite fields do not vanish identically on a nonempty open unless zero.

Lemma 0.1. *Let A be a ring. Let k be a field. Let $\mathfrak{p}_n$, $n \geq 1$ be a sequence of pairwise distinct primes of A. Moreover, for each n let $k \to \kappa(\mathfrak{p}_n)$ be an embedding. Then the closure of the image of*

$$\coprod\nolimits_{n \neq m} \operatorname{Spec}(\kappa(\mathfrak{p}_n) \otimes_k \kappa(\mathfrak{p}_m)) \longrightarrow \operatorname{Spec}(A \otimes A)$$

meets the diagonal.

Proof. Set $k_n = \kappa(\mathfrak{p}_n)$. We may assume that $A = \prod k_n$. Denote $x_n = \operatorname{Spec}(k_n)$ the open and closed point corresponding to $A \to k_n$. Then $\operatorname{Spec}(A) = Z \amalg \{x_n\}$ where Z is a nonempty closed subset. Namely, $Z = V(e_n; n \geq 1)$ where e_n is the idempotent of A corresponding to the factor k_n and Z is nonempty as the ideal generated by the e_n is not equal to A. We will show that the closure of the image contains $\Delta(Z)$. The kernel of the map

$$(\prod k_n) \otimes_k (\prod k_m) \longrightarrow \prod\nolimits_{n \neq m} k_n \otimes_k k_m$$

is the ideal generated by $e_n \otimes e_n$, $n \geq 1$. Hence the closure of the image of the map on spectra is $V(e_n \otimes e_n; n \geq 1)$ whose intersection with $\Delta(\operatorname{Spec}(A))$ is $\Delta(Z)$. Thus it suffices to show that

$$\coprod\nolimits_{n \neq m} \operatorname{Spec}(k_n \otimes_k k_m) \longrightarrow \operatorname{Spec}(\prod\nolimits_{n \neq m} k_n \otimes_k k_m)$$

has dense image. This follows as the family of ring maps $\prod_{n \neq m} k_n \otimes_k k_m \to k_n \otimes_k k_m$ is jointly injective. □

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.